

Finite Mechanism Structure

Recursive Return in Single-Defect Circular Automata

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This paper (Paper XXVIII of the RIME program) develops finite mechanism structure, exact credit composition, and fixed-scope recursive return in single-defect circular automata. It inherits the typed entry and completion interface of Paper XXVII without enlarging or re-owning the theorem scope established there.

Abstract

Problem. Exact descent relations contain more information than a mechanism label. Packet identities, typed boundaries, ancestry, and the allocation of maturity credit can all affect semantic composition and recursive return.

Approach. We retain exact realization fibers on declared finite carriers of single-defect circular automata, separate interaction skeletons from accounting refinements, and compare five independently constructed return surfaces through typed handoff and provenance-sensitive mechanism predicates.

Results. On a closure of four rank-four contexts at $n = 6$ and thirty-five at $n = 7$, the accounting and skeleton quotients fail to be semantic composition congruences. A boundary-relational description factors 11,252 exact receipts into transport, return, and fusion operations. The credit identity $B(C) = L(x) + S(x) + B(C')$ composes on exact typed lifts and holds on all 13,054 compatible receipt pairs. Five declared $n = 7$ carriers support existential return from rank five through rank four to the exact low-rank base. Their 165 tagged rank-four contexts admit a sufficient GFPC/PEC mechanism cover, and all 26,995 macro lifts align with canonical one- or two-segment first-exit paths.

Boundary. These are finite-structure results, not an all-rank constructor, a uniform mechanism selector, or a reset bound. Projectable-Origin Supply—the existence of a projectable member in every admitted inherited transfer fiber—remains open at arbitrary ambient size.

Keywords. synchronizing automata; single-defect circular automata; typed descent; exact realization fibers; credit composition; recursive return; computational certificate

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}, p, d$	state set, labelled cycle, and binary-kernel rank- $(n - 1)$ defect
$C = (A, d, \mu_r, \chi_r)$	certified rank- r source context
$\text{Lift}_r(C, m)$	complete exact realization fiber of declared menu channel m
$\widehat{\partial^+ x}$	theorem-facing typed endpoint after verified handoff
$P_{\leq 3}$	exact nonrecursive low-rank base
B, L, S	credit budget, exact word length, and corridor surplus
GFPC, PEC	Generalized Fresh-Pair Carry and Pair-Extension Carry
$\mathcal{T}_{5 \rightarrow 4}(z)$	complete declared transfer fiber of inherited pair z
$\mathcal{T}_{5 \rightarrow 4}^{\text{proj}}(z)$	projectable subfiber, not a chosen transfer

1 Introduction

A recursive descent argument needs both a legal proof state and a realization that reaches another legal proof state. The first requirement does not determine the second. Two paths can share their fusion pattern and numerical credit data while exposing different typed successor relations. Conversely, distinct exact paths can support the same finite return theorem without sharing a shortest word, a surplus value, or a literal target serialization.

This paper studies the finite structure between these two levels. We use the source-addressed proof interface of Paper XXVII [3] and retain three objects separately:

interaction skeleton + accounting refinement + exact typed realization fiber.

Our contributions are organized by their mathematical dependencies.

1. On a declared $4 + 35$ seed closure, we locate the failure of quotient-level semantic composition and give an exact boundary-generator cover.
2. We prove the exact credit telescope and record its finite, relation-valued accounting composition.
3. We combine five fixed- $n = 7$ return surfaces into one existential section-to-section-to-base statement, retaining unsuccessful channels.
4. We give a reduced sufficient GFPC/PEC cover of 165 tagged rank-four contexts and transport it through canonical first-exit path alignment.

The finite domains are part of the statements. The $4 + 35$ closure used for composition is not the 165-context domain used for the five returns. Nor is either domain an enumeration of all admitted contexts at arbitrary ambient size. The computational certificates retain complete tied fibers within their declared constructions; they are not lists of selected winners.

The final question concerns the origin of a recursive input rather than a new finite realization: what all- n structure forces an admitted inherited pair to have a projectable origin? The finite results below do not depend on a solution to that question.

1.1 Related Work and Novelty Boundary

For circular automata, Dubuc proved the Černý bound in a broader circular class [6]. One-cluster arguments provide a different route under their own cycle hypotheses [7]. Neither result is reproved or improved here: our scope is a finite collection of typed rank-descent relations, not a uniform reset-length theorem.

The alphabet of a permutation together with a rank- $(n - 1)$ merging letter also occurs in work on randomized construction of slowly synchronizing automata [2]. Completely reachable

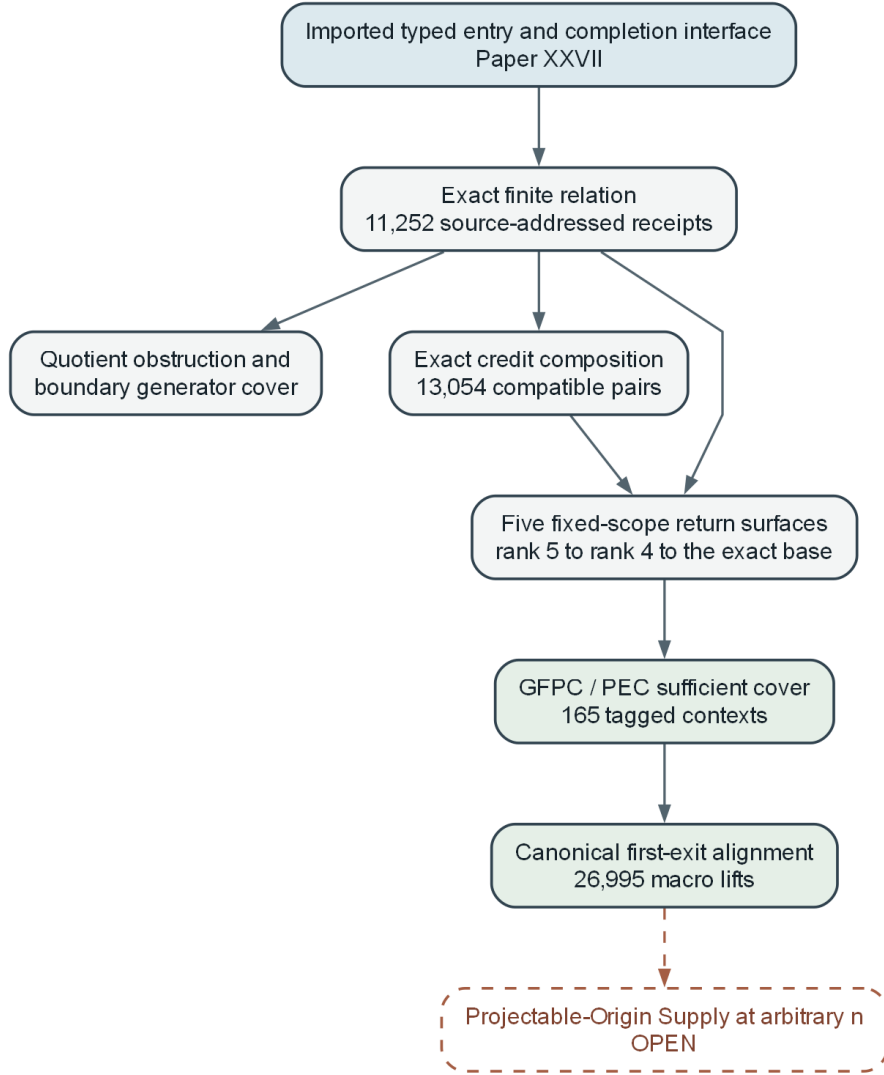


Figure 1. Finite proof architecture. Solid arrows denote construction dependencies within the declared finite scopes. The dashed boundary records an open all- n supply problem, not an implication from the finite certificates.

automata ask whether every nonempty subset can arise as an image [1]. We assume neither that reachability property nor a randomized generator model. Packet identity, proof-context certification, and tied exact realizations are retained because the recursive return question reads them.

Within this series, Paper XXIII supplies deterministic pair-hitting and marked-kernel corridor geometry [4], whereas Paper XXVI treats stochastic pair transfer and waiting [5]. Paper XXVII owns the fixed-scope entry section and source-addressed completion interface used here [3]. The present contribution is the finite mechanism quotient obstruction, exact credit composition, five relation-valued recursive returns, a reduced sufficient tagged cover, and canonical path alignment. Paper XXVII asks which proof information must be retained; this paper asks how exact descent propagates within that retained interface. These results neither re-own the imported interface nor establish Projectable-Origin Supply at arbitrary n .

2 Imported Proof Interface

Let $Q = \mathbb{Z}/n\mathbb{Z}$, let p be the labelled cyclic permutation, and let d be a labelled rank- $(n-1)$ defect with a binary kernel. Words act from left to right. An exact state retains source-addressed atom packets and their occupied coordinates, not only the packet masses.

A certified rank- r context is written

$$C = (A, d, \mu_r, \chi_r).$$

Here the characteristic χ_r carries the distinguished and ancestry data required by the owning theorem. A legal checkpoint, an internal replay state, and a certified lower target are different types of objects. Paper XXVII supplies the checkpoint interface, source-addressed completion relations, and endpoint-normalization and accounting gates. Those objects are imported without reproof. The typed handoffs used below are declared and validated by the five return constructions of this paper.

An exact lift x has source ∂^-x , actual endpoint ∂^+x , and theorem-facing target

$$\widehat{\partial^+x} = \eta_x(\partial^+x).$$

The notation includes the declared normalization semantics. The handoff may be identity or a verified typed atom bijection, but it must preserve every observable used by the lower theorem. Equality of mass partitions alone does not provide a handoff.

For a declared source C , a finite menu $\mathfrak{M}_r(C)$ has relation-valued exact fibers $\text{Lift}_r(C, m)$. We retain the following construction requirements from the paper-owned return schema:

- **F1, future-free construction.** The menu, channel keys, and lift relation are constructed without access to lower-section membership, low-rank success, winning labels, or reset coaccessibility.
- **F2, source typing.** Every lift in $\text{Lift}_r(C, m)$ has source C .
- **F3, exact soundness.** Replay, packet roles, ancestry, normalization, handoff, and accounting satisfy the declared typed equations.
- **F4, checkpoint typing.** Internal replay states are not exported as recursive checkpoints without independent certification.

For a separately certified lower destination class $\mathfrak{D}_{<r}$, define

$$\text{Good}_r(C, m) \iff \exists x \in \text{Lift}_r(C, m) : \widehat{\partial^+x} \in \mathfrak{D}_{<r}. \quad (2.1)$$

Thus success is evaluated on the constructed relation; it is not part of its construction. The return assertion, F5, is

$$\forall C \in \text{Sec}_r \quad \exists m \in \mathfrak{M}_r(C) : \text{Good}_r(C, m). \quad (2.2)$$

F1–F4 do not imply F5. Sections 5 and 6 establish finite instances, not an all-rank realization of these contracts. Per-context finiteness of a menu also does not imply a rank-controlled or an absolute uniform size bound.

3 Mechanism Quotients and Boundary Generators

3.1 The exact finite relation

Let $\mathcal{R}_{\text{seed}}$ be the exact relation reconstructed from the four declared $n = 6$ rank-four contexts and the thirty-five declared extremal $n = 7$ contexts, together with their strictly descending rank-three and rank-two closure. The construction retains every tied endpoint-shortest admissible receipt in its declared Type-I/II relation. Type I uses one corridor with nonnegative surplus; Type II uses two corridors with $S_1 < 0$ and $S_1 + S_2 \geq 0$. These are local accounting conditions, not membership tests in the lower success set.

The paper-owned seed certificate has the following scope.

Object	Count
Rank-four seed contexts	39
Seed receipts	4,250
Lower-rank closure receipts	7,002
All exact receipts	11,252
Exact compatible receipt pairs	13,054
Type-I receipts	1,634
Type-II receipts	9,618

An exact receipt retains its source, word, corridor boundaries, fusion packet identities, endpoint, and ancestry update. Source, endpoint, and total length alone are not an exact-receipt identity.

The accounting quotient q_{acct} retains the interaction skeleton, rank drops, maturity gains, corridor lengths and surpluses, debt profile, target partition, and residual budget. The skeleton quotient q_{skel} forgets the numerical allocation while retaining the declared source/target partitions, fusion pattern, and interaction/ancestry-participation type. Write q_{skel} also for its composite with q_{acct} . The complete key definitions are retained in the paper-owned evidence package.

For a quotient q , an observable O descends if

$$q(x) = q(x') \implies O(x) = O(x').$$

For set-valued observables this is equality of the full images, not agreement of one chosen representative.

3.2 Failure of semantic composition congruence

Exact compatibility $\text{Comp}(x, y)$ requires equality of the typed target of x and source of y , including the required ancestry. Define

$$\text{Succ}_q(x) = \{q(y) : \text{Comp}(x, y)\}.$$

A semantic composition congruence would require

$$q(x) = q(x') \implies \text{Succ}_q(x) = \text{Succ}_q(x'). \quad (3.1)$$

Proposition 3.1 (Finite quotient obstruction; computational certificate).

On $\mathcal{R}_{\text{seed}}$, neither the accounting quotient nor the skeleton quotient satisfies (3.1). Even nonemptiness of the successor image fails to descend.

Quotient	Source fibers	Noncongruent fibers	Successor- nonemptiness mismatches
Accounting	250	103	69
Skeleton	30	9	8

Certificate proof. The seed reconstruction forms compatibility edges from exact typed boundary equality and compares the complete successor-class sets in each quotient fiber. The displayed noncongruent fibers contain explicit unequal images; the last column records unequal emptiness status. The same certificate separates the unary observables: ranks, partitions, fusion mass patterns, and the residual budget descend to the skeleton; lengths, surpluses, and debt require accounting; exact packet identities, ancestry updates, and target channels still require the exact relation. Neither quotient uses downstream success as a key.

The boundary audit verifies, on this same finite closure, that the labelled action and normalized target mass placement support fiber-uniform accounting/skeleton successor classes. This is a finite liftability result, not an all-rank boundary congruence. In particular, the obstruction does not justify replacing exact compatibility by equality of accounting tuples.

3.3 A boundary-relational generator cover

For each exactly replayed operation use a row

$$b^- \xrightarrow{g,a} b^+.$$

The abstract boundaries retain the action and normalized mass placement; a records length, maturity gain, surplus, debt, and budgets. Its realization fiber retains the exact packet placement, distinguished packet, word segment, and receipt/corridor/operation addresses. Composition still requires a compatible exact intermediate state.

Split each stored corridor at maximal nonempty permutation blocks and defect letters:

Generator	Segment	Rank behavior
TRANSPORT	p^a , $a > 0$, maximal within the corridor	Rank preserving
RETURN	d	Rank preserving
FUSION	Terminal d	One binary strict drop

Proposition 3.2 (Exact generator cover; computational certificate).

Every receipt of $\mathcal{R}_{\text{seed}}$ is covered by these typed operation rows, with its endpoint and accounting recovered by exact replay. There are 123,240 operation occurrences:

$$49,656 \text{ TRANSPORT} + 52,714 \text{ RETURN} + 20,870 \text{ FUSION}.$$

Grouping by (b^-, g, a, b^+) gives 24,355 rows, each with a nonempty exact realization fiber.

Generator	Relation rows	Largest exact fiber
TRANSPORT	12,053	200
RETURN	8,113	336
FUSION	4,189	192

Certificate proof. Split each stored exact word as above and replay every segment from its source-addressed state. Each corridor has one terminal fusion; the recomposed word, endpoint, length, surplus, and credit allocation agree with its receipt. Every row is populated by its recorded occurrences, and the uncovered receipt set is empty.

The split is canonical for a fixed word, not unique among tied exact words. Nor does nonemptiness of these rows assert realizability of every formally writable row outside this finite relation.

This description separates interaction operations from path predicates. Repayment records a credit condition; a heavy chain records consumption of the first fresh packet by the second fusion. There are 9,618 repayment receipts but only 9,608 repeated-fusion heavy chains. Fallback is a target/allocation choice, not a fourth interaction operation. The resulting object is a typed compatibility graph with accounting decoration and exact fibers, not an algebra on three bare generator names.

4 Exact Credit Composition

4.1 Maturity and the telescope

For a mass state μ of total mass n , let

$$r(\mu) = |\text{supp } \mu|, \quad M(\mu) = \sum_q \binom{\mu(q)}{2}.$$

Use the maturity potential of Paper XXVII:

$$\tau(\mu) = \|\mu\|_2^2 + (r - 1)(n - r) - 2n + 1$$

for nonuniform states, with the separate initial convention $\tau(\mathbf{1}) = 0$. Write $\tau(n)$ for the rank-one reset maturity. For a typed context C , set

$$B(C) = \tau(n) - \tau(\mu(C)).$$

If an exact corridor has length ℓ and endpoints of ranks $r, r - q$, its surplus is

$$S = \tau(\mu') - \tau(\mu) - \ell = 2\Delta M + q(2r - q - n - 1) - \ell. \quad (4.1)$$

The second expression uses the nonuniform-state formula on the corridor's endpoints. For an exact receipt $x : C \rightarrow C'$, sum its corridor lengths and surpluses to obtain $L(x)$ and $S(x)$. When $C' = \widehat{\partial^+ x}$, the declared handoff is either the identity or a mass-preserving typed atom bijection. Hence the target mass state is only relabelled and

$$\tau(\mu(C')) = \tau(\mu(\partial^+ x)). \quad (4.2)$$

Proposition 4.1 (Exact credit composition).

Every such receipt satisfies

$$\boxed{B(C) = L(x) + S(x) + B(C')}. \quad (4.3)$$

For exact compatible receipts $x : C_0 \rightarrow C_1$ and $y : C_1 \rightarrow C_2$,

$$\begin{aligned} L(y \circ x) &= L(x) + L(y), \\ S(y \circ x) &= S(x) + S(y), \\ B(C_0) &= L(y \circ x) + S(y \circ x) + B(C_2). \end{aligned} \quad (4.4)$$

Proof. Summing (4.1) cancels the intermediate maturity values, so $S(x) = \tau(\mu(C')) - \tau(\mu(C)) - L(x)$, which is (4.3). Exact compatibility supplies the same middle context to both identities. Adding them cancels $B(C_1)$ and proves (4.4). This proves the numerical identity on an existing exact composition; it does not produce that composition or assert membership of the concatenation in a shortest fiber.

4.2 Accounting fibers and allocation

Proposition 4.2 (Finite accounting composition; computational certificate).

All 11,252 receipts and all 13,054 exact compatible pairs in the seed closure satisfy (4.3)–(4.4). There are 569 accounting-label pairs admitting an exact compatible lift. Within each such pair fiber, the composite accounting summary is constant: partition chain, gross credit, length and surplus vectors, tail budgets, debt profile and peak debt, and final allocation.

Certificate proof. For each compatible exact pair, compute both middle budgets independently and form the composite summary. Group these summaries by the accounting-label pair and test constancy of the full image. Of the 569 pair fibers, 507 have more than one exact lift, and one has 616. Thus constancy is not uniqueness of the concrete path. The paper-owned credit certificate reconstructs these identities and fibers.

This defines a partial, exact-lift-supported accounting composition:

$$(a_1, a_2) \xrightarrow{\exists \text{ compatible exact lift}} A_{21}.$$

It is neither a total product nor an inverse test for lift existence. Proposition 3.1 explains why the accounting result cannot determine semantic composability.

The 26 realized skeleton-label pairs provide a sharper contrast.

Composite observable	Skeleton pair fibers where it varies
Partition chain, gross credit, or tail-budget vector	0
Length or surplus allocation	23
Debt profile or peak debt	15

The skeleton retains gross geometry, but not its allocation. For example, the extremal rank-four source has budget 27. Its zero-surplus subfamilies include 1,020 exact heavy-target receipts with prefix/tail allocation $20 + 7$, and 87 fallback receipts with allocation $12 + 15$:

$$20 + 7 = 12 + 15 = 27.$$

The fallback creates no credit. It changes where the existing credit is spent. Neither this equality nor the additive telescope decides whether a candidate word belongs to an admissible lift relation.

5 Five Fixed-Scope Recursive Return Surfaces

5.1 Carriers and future-free relations

Let $i \in I = \{\text{ext}, 2, 3, 4, 5\}$. The symbols $\text{Sec}_{5,i}^{(7)}$ and $\text{Sec}_{4,i}^{(7)}$ denote the five separately declared finite carriers and their certified checkpoint status. The first rank-five carrier, called $\text{Sec}_{5,\text{cand}}^{(7)}$ in its certificate, is indexed here by ext to match its rank-four destination. Tags are retained. Contexts with equal projections remain distinct across carriers.

Each carrier is specified by its source-local selector and complete candidate payload, not by its successful lifts or by the descriptive table below. The five declaration/result pairs are denoted ext, cand2, cand3, cand4, and cand5. Appendix A identifies the paper-owned evidence package; its index binds each pair to the corresponding rank-four and rank-five records.

For each declared source the construction forms the complete tied exact relation in its stated scope, groups it into future-free channels, and retains all accounting refinements and exact lifts. It has no access to lower-section membership. Target certification applies to this fixed relation.

Consequently unsuccessful lifts and whole failed channels remain part of the mathematical object.

The successful rank-five transfers have the following anatomy. The two tables separate carrier size and accounting from provenance-sensitive transport.

Realization	$ \text{Sec}_5 $	$ \text{Sec}_4 $	Terminal fusion	Surplus	Length
ext	35	35	1 + 1	0	3
cand2	48	48	1 + 1	0	3
cand3	36	36	1 + 2	2	3
cand4	36	36	1 + 2	2	3
cand5	10	10	1 + 2	0	5

Realization	Incoming distinguished packet	Typed handoff
ext	Not consumed	Verified atom bijection
cand2	Not consumed	Identity
cand3	Not consumed	Identity
cand4	Consumed	Identity
cand5	Consumed	Identity

The ext source/target mass partitions are $(2, 2, 1, 1, 1)$ and $(2, 2, 2, 1)$. For cand2 they are $(3, 1, 1, 1, 1)$ and $(3, 2, 1, 1)$; for cand3–5 they are $(2, 2, 1, 1, 1)$ and $(3, 2, 1, 1)$. These partitions do not by themselves define the carriers.

5.2 The finite return theorem

Theorem 5.1 (Five fixed-scope return chains; computational certificates).

For every $i \in I$, the declared future-free menu/lift constructions satisfy F1–F4 and the following existential return statements:

$$\begin{aligned} \forall C_5 \in \text{Sec}_{5,i}^{(7)} \exists m_5 \in \mathfrak{M}_{5,i}(C_5) \exists x \in \text{Lift}_{5,i}(C_5, m_5) : \widehat{\partial^+ x} \in \text{Sec}_{4,i}^{(7)}, \\ \forall C_4 \in \text{Sec}_{4,i}^{(7)} \exists m_4 \in \mathfrak{M}_{4,i}(C_4) \exists y \in \text{Lift}_{4,i}(C_4, m_4) : \widehat{\partial^+ y} \in P_{\leq 3}^{(7)}. \end{aligned} \quad (5.1)$$

Here $P_{\leq 3}^{(7)}$ is the imported exact low-rank base. Hence, with arrows interpreted existentially through typed handoff,

$$\boxed{\text{Sec}_{5,i}^{(7)} \longrightarrow \text{Sec}_{4,i}^{(7)} \longrightarrow P_{\leq 3}^{(7)}} \quad (5.2)$$

Certificate proof. The source declarations and relation constructors fix each candidate domain independently of its return evaluator. Exact replay validates the source, fusion, ancestry, normalization and handoff fields of each lift. The rank-four certificates establish a successful channel for every declared rank-four source. The rank-five certificates establish a lift into that independently certified lower carrier for every declared rank-five source. Compose these existential statements using the same certified handoff of the chosen rank-five lift. Intermediate replay states do not acquire checkpoint status. The complete finite counts are as follows.

Realization	Rank-five channels	Rank-five lifts	Returning channels	Returning lifts
ext	87	214	35	35
cand2	96	195	48	48
cand3	70	209	36	36
cand4	107	252	36	36
cand5	28	62	10	10

Realization	Rank-four channels	Rank-four lifts	Returning channels	Returning lifts
ext	173	4,182	169	2,329
cand2	401	9,600	401	9,498
cand3	172	5,239	171	4,208
cand4	252	6,498	252	5,937
cand5	53	1,476	53	1,205
Tagged total	1,051	26,995	1,046	23,177

A returning channel means one with at least one returning lift, not a channel all of whose lifts return. At rank four the relation retains 3,818 unsuccessful lifts, five entirely unsuccessful channels, and 186 mixed channels. The ext handoffs are all verified nonidentity atom bijections; the other four surfaces use identity handoffs. These checks establish (5.1) on the stated carriers and preserve its existential quantifiers. See the paired source and evaluation certificates in Appendix A.

5.3 Common structure and non-necessary anatomy

Corollary 5.2 (Limits of a common finite normal form).

None of the following is a necessary feature shared by the five declared successful rank-five return surfaces: 1 + 1 fusion, zero surplus, length three, avoidance of the incoming distinguished packet, or literal identity of the handoff.

Proof. *Cand3–5 use 1 + 2 fusion; cand3 and cand4 have surplus two; cand5 has successful length five; cand4 and cand5 consume the incoming distinguished packet; ext uses nonidentity handoffs. Each statement is witnessed within Theorem 5.1’s independently declared finite domains.*

The invariant requirements are typed, source-addressed realization and valid credit/handoff, not these particular anatomical values. This corollary does not assert that every alternative anatomy is realizable. Length and surplus also obey the same credit equation: the cand4/cand5 comparison is not an experiment varying them independently.

The scope is intentionally narrower than universal return. Theorem 5.1 does not prove success of every lift, maximality of any carrier, a rank-controlled menu bound, or a constructor on arbitrary n, r .

6 Reduced Mechanism Cover and Canonical Alignment

6.1 Two provenance-sensitive mechanism families

A mechanism support predicate reads an exact selected-entry provenance κ_4^{ISE} , not only the final mass partition. It includes the upstream packets $\Pi^\uparrow$, selected entry e , its terminal fresh packet F_{ent} , source certification, ancestry update, and typed handoff η into the rank-four context C .

The two sufficient families have the following source-addressed meanings.

Generalized Fresh-Pair Carry (GFPC). There are five upstream packets and four target packets. Two singleton packets x, y undergo the terminal strict fusion $x \sqcup y = F_{\text{ent}}$. The three other packets carry atomwise, including the incoming distinguished packet, and

$$\eta(F_{\text{ent}}) = \text{Dist}(\chi_4), \quad \Pi_C = \eta((\Pi^\uparrow \setminus \{x, y\}) \cup \{F_{\text{ent}}\}). \quad (6.1)$$

The branch does not fix the background mass partition or require a literal identity handoff.

Pair-Extension Carry (PEC). There are distinct upstream packets P, Q, x , with $|P| = |Q| = 2$ and $|x| = 1$. The terminal strict fusion is $P \sqcup x = F_{\text{ent}}$, the other pair Q carries atomwise, and

$$\eta(F_{\text{ent}}) = \text{Dist}(\chi_4), \quad \Pi_C = \eta((\Pi^\uparrow \setminus \{P, x\}) \cup \{F_{\text{ent}}\}). \quad (6.2)$$

The incoming distinguished packet is either P or Q , with its consumed-or-carried update certified exactly. The handoff preserves the fresh, carried-pair, and spectator roles.

Both predicates retain their role witnesses relationally. Neither reads the selected word, length, surplus, lower success, or a winner. The historical Fresh-Pair Carry predicate is a specialization of GFPC at the branch-domain level; this inclusion does not transfer a completion theorem to the larger domain.

6.2 Reduced finite mechanism cover

Let $\mathcal{D}_4^{(7)}$ be the tagged disjoint union of the admitted rank-four context/certificate pairs of the five carriers. Thus

$$|\mathcal{D}_4^{(7)}| = 35 + 48 + 36 + 36 + 10 = 165.$$

For $g \in \{\text{GFPC}, \text{PEC}\}$, let $\mathcal{G}_g^{\text{fs}}(z)$ be the declared fixed-scope relation of supported provenance and exact local-return witnesses for z . A support witness and a component-completion witness are both required; they are not the same predicate.

Theorem 6.1 (Reduced finite mechanism cover; computational certificate).

For every $z \in \mathcal{D}_4^{(7)}$,

$$\boxed{(\{\text{GFPC}\} \times \mathcal{G}_{\text{GFPC}}^{\text{fs}}(z)) \sqcup (\{\text{PEC}\} \times \mathcal{G}_{\text{PEC}}^{\text{fs}}(z)) \neq \emptyset.} \quad (6.3)$$

GFPC supplies ext and cand2; PEC supplies cand3, cand4, and cand5.

Certificate proof. The exact provenance records validate (6.1) on $35 + 48 = 83$ tagged contexts and (6.2) on $36 + 36 + 10 = 82$. The separate GFPC and PEC component certificates verify generic LocalReturn on their respective complete finite adapters, retaining failed and mixed channels. Their tagged composition covers all 165 contexts. Its counts are the rank-four totals in Theorem 5.1, with 3,162 distinct certified tagged targets. No new return evaluation or winner choice enters this composition.

This is a reduced **sufficient** cover. It is not a proof that two families are minimal or unique, that support alone implies completion, or that the same families cover every admitted all- n context.

6.3 Alignment with canonical first-exit paths

For a typed context C , the canonical relation $\Lambda_4^{\text{complete}}(C)$ uses the declared rank-four plateau and first strict exit, complete typed endpoint normalization, endpoint-shortest distance, and all tied shortest realizations. Its construction does not read lower success. The analogous rank-three relation is $\Lambda_3^{\text{complete}}$.

The finite adapter $\Lambda_4^{\text{fs}}(C)$ is a different relation type: it contains Type-I/II macro receipts. A Type-I receipt has one strict-exit segment. A Type-II receipt has two, with the intervening debt compatibility. Removing artifact-only identities from a receipt does not remove its words, packet identities, corridor boundaries, ancestry, or accounting.

Theorem 6.2 (Canonical alignment; computational certificate).

On $\mathcal{D}_4^{(7)}$, every finite adapter macro lift has an explicit canonical first-exit path image:

$$\boxed{\Lambda_4^{\text{fs}}(C) \hookrightarrow \text{Path}_{1,2}(\Lambda_4^{\text{complete}}, \Lambda_3^{\text{complete}}).} \quad (6.4)$$

The alignment preserves typed boundary, ancestry, normalization, exact observables, and ISE provenance. It transports Theorem 6.1’s supported return witnesses to canonical certificates. The path-image map is injective on each declared tagged adapter relation: no two distinct typed macro lifts have the same typed one- or two-segment path image.

Carrier	Sources / transfers	Canonical Λ_4 rows	Adapter macro paths
ext	35 / 35	1,537	4,182
cand2	48 / 48	4,178	9,600
cand3	36 / 36	3,511	5,239
cand4	36 / 36	3,284	6,498
cand5	10 / 10	939	1,476
Total	165 / 165	13,449	26,995

Certificate proof. Reconstruct the canonical first-exit relations independently of the macro adapters. Exact replay identifies each adapter receipt with its one-segment or debt-compatible two-segment image, retaining all theorem-facing fields. There are 1,528 Type-I and 25,467 Type-II images. The source-boundary and provenance checks hold on all 165 contexts. Support transports by the same ISE provenance records; LocalReturn transports by the exact path image. The 23,177 already certified returning macro lifts therefore supply canonical returning paths. The alignment certificate binds the construction independently of the component-success data.

The conclusion is alignment, not $\Lambda_4^{\text{fs}}(C) = \Lambda_4^{\text{complete}}(C)$. A two-segment macro is not a single first exit, and neither mathematical relation equality nor raw serialization equality follows from (6.4). This distinction permits semantic transport without erasing the types of the compared objects.

7 Boundaries and Projectable-Origin Supply

7.1 The finite boundary

The quotient obstruction, generator cover, and finite accounting constancy refer to the $4 + 35$ closure. The return, reduced-cover, and alignment theorems refer to the five tagged $n = 7$ carriers. The algebraic telescope applies whenever its exact typed composition premises hold, but it supplies neither admissibility nor recursive checkpoint status.

7.2 The remaining origin question

Fix the declared canonical, unrefined interfaces and an admitted pair $z = (C, \kappa_4^{\text{ret}})$, where

$$\kappa_4^{\text{ret}} = (b_4, \chi_4, \Lambda_4^{\text{complete}}(C), \nu_4).$$

Its complete transfer fiber is

$$\mathcal{T}_{5 \rightarrow 4}(z) = \{\tau : \text{TransferRecord}_5(\tau), \text{tgt}(\tau) = C, \text{ret}(\tau) = \kappa_4^{\text{ret}}\}.$$

Completeness is relative to this declared relation. It means neither arbitrary exact-valid words nor a producer-selected sublist. Let

$$\begin{aligned} \text{Sec}_4^{\text{inh}} &= \{z \in \text{Sec}_4 : \mathcal{T}_{5 \rightarrow 4}(z) \neq \emptyset\}, \\ \mathcal{T}_{5 \rightarrow 4}^{\text{proj}}(z) &= \{\tau \in \mathcal{T}_{5 \rightarrow 4}(z) : \exists \kappa_4^{\text{ISE}} \text{ Proj}_{\text{ISE}}(\tau, \kappa_4^{\text{ISE}})\}. \end{aligned} \tag{7.1}$$

Projection requires an actual compatible selected-entry provenance, exact replay, ancestry update, and typed target handoff on the same transfer. The inherited domain is not defined by projectability.

Open Problem 7.1 (Projectable-Origin Supply). Under the declared relation semantics and their applicability premises, does

$$\boxed{\forall z \in \text{Sec}_4^{\text{inh}}, \quad \mathcal{T}_{5 \rightarrow 4}^{\text{proj}}(z) \neq \emptyset} \quad (7.2)$$

hold at arbitrary ambient size?

The 165 finite fibers audited above are projectable singletons. They establish (7.2) on that finite domain, but cannot distinguish existential supply from the stronger Uniform Projection assertion $\mathcal{T}^{\text{proj}}(z) = \mathcal{T}(z)$. Uniform Projection is not needed in (7.2).

An entry-evidence condition together with projection soundness for the same (τ, e, sel) would be a sufficient factorization of this goal. Neither all- n premise is established here. In particular, entry evidence alone is not a projection witness. Equation (7.1) fixes the complete fiber and (7.2) fixes the existential quantifier. Both the entry and the projection must be witnessed by the same transfer; an entry witness for one transfer cannot be combined with a projection calculation for another.

A counterexample to (7.2) must give an admitted z with

$$\emptyset \neq \mathcal{T}_{5 \rightarrow 4}(z), \quad \mathcal{T}_{5 \rightarrow 4}^{\text{proj}}(z) = \emptyset.$$

It must exclude every projection of every member of that fixed complete fiber. One unsuccessful transfer, a failed proof route, or an unspecified predicate does not meet this standard. No qualified whole-fiber counterexample is established here.

A proof of (7.2) would still not supply a uniform menu, an all-rank cover/admission theorem, or an all-rank recursive selector. Those are separate open structural problems, not conclusions of the finite certificates.

The finite mechanism structure, credit calculus, five recursive returns, reduced cover, and canonical alignment constitute the finite results. The unresolved all- n question is which additional structure forces a projectable recursive origin.

8 Claim Status and Boundary

Claim surface	Evidence level	Bound
Credit telescope and formal composition identity	Theorem	Exact typed lifts under the stated accounting equations
Seed quotient obstruction, finite generator cover, five returns, GFPC/PEC cover, and canonical alignment	Computational Certificate	The declared $n = 6/n = 7$ source relations and their receipt-bound finite closure
Projectable-origin supply, uniform menus, all-rank cover/admission, and an all-rank recursive selector	Research Program	No all- n proof or qualified whole-fiber counterexample

The finite certificates are reproducible checks of declared complete relations, not statistical observations. They do not certify unenumerated sources, a minimum mechanism family, equality of macro and first-exit relations, a universal reset bound, or independent implementation agreement. The paper-owned evidence package preserves the original artifacts and receipts byte-for-byte; its separate validation receipt certifies local release-copy closure only.

9 Conclusion

Exact realization fibers cannot in general be replaced by bare mechanism labels or accounting tuples. On the declared finite carriers, the relevant structure is instead a typed relation: boundary operations retain concrete realizations, credit composes only along compatible lifts, and recursive return is existential over independently constructed finite menus.

The five return surfaces nevertheless admit a common sufficient description. GFPC and PEC cover all 165 tagged rank-four contexts, and their 26,995 macro lifts align with canonical one- or two-segment first-exit paths. This closes the finite mechanism analysis without promoting the cover to a universal selector. The remaining all- n problem is Projectable-Origin Supply.

Appendix A Computational Artifacts and Certificate Scope

The finite claims above use exact replay and source-addressed finite relations, not statistical sampling. Algebraic Proposition 4.1 has a direct proof; a supplementary Lean spine checks its conditional credit algebra and the data-independent composition and transport implications used by Theorems 5.1 and 6.2. The finite counts, fiber constancy, return coverage, and alignment statements retain their separate computational certificates. No claim of an independent second implementation is made.

The manuscript is self-contained at its stated claim levels. The accompanying source-addressed evidence package is available in the [RIME repository](#) under `experiments/paper28/`; these paths are reproducibility pointers, not premises needed to read the proofs.

Artifact surface	Role	Repository-relative short path
Theorem-facing records	finite relations, return surfaces, cover, and alignment	<code>artifacts/</code> , <code>results/</code>
Producers	exact finite construction and replay	<code>*.py</code>
Validators	scientific checks and receipt validation	<code>validation/</code>
Dependency manifest	ordered frozen mirror and exact input graph	<code>dependency-manifest.json</code>
Finite validation receipt	replay of the 20 finite roots; downstream of the historical receipts	<code>validation/</code>
Supplementary Lean spine	conditional algebra, composition, and transport implications	<code>lean/</code> , <code>results/</code>
Release manifest	manuscript/build roots and the two nested evidence closures	<code>release-manifest.json</code>
Public package receipt	outer local closure verification; excluded from every upstream closure	<code>results/</code>

Each result package supplies its exact source declarations, relation constructor, artifact, receipt, and validator. The package README lists the file-level bindings and validation commands. Candidate payloads define the finite domains; the anatomy table in the main text is not a replacement for those declarations.

The evidence index records exact digests, producers, validators, commands, and the two primary immutable artifact hashes. The paper-owned local validation receipt checks the frozen mirror and replays the 20 manuscript-level finite validators without an external source tree. It is downstream of the original receipts and does not re-sign them or supply independent implementation agreement. The construction/evaluation separation preserves failed rows and prevents lower success from becoming a hidden channel key.

The outer release manifest binds the manuscript, reader PDF, bibliography, figure sources, finite mirror, Lean receipt, and public-package validator. Its downstream public receipt is not a member of the manifest, either nested closure, release identity, or package inventory.

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